7 Answer: C

Since the balloon is in equilibrium,

upthrust on balloon = weight of balloon and helium + F_s cos30

$$\rho_{air} V_{balloon} g = \rho_{helium} V_{balloon} g + m_{balloon} g + ke \cos 30$$

$$(1.29)(4.50)g = (0.180)(4.5)g + \left(\frac{15}{1000}\right)g + (80)e\, cos30$$

$$56.94705 = 7.9461 + 0.14715 + 69.282\, e$$

$$e = 0.705 \text{ m}$$